



SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences

Chennai- 602105



Student Names:

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Course Code: MLA0408

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Course Name: Deep Learning

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Course Faculty: Dr. D. Sudhagar

Title: CNN-Based Traffic Sign Recognition System

Assignment Description:

This assignment focuses on developing a **Traffic Sign Recognition System using Convolutional Neural Networks (CNNs)**. The objective is to automatically classify different categories of traffic signs by following the complete deep-learning workflow, including dataset preparation, image preprocessing, CNN architecture design, training, testing, and performance evaluation. A suitable subset of the **GTSRB dataset** is used to maintain manageable training time in Google Colab. Images are resized and normalized, while augmentation techniques such as rotation, shifting, and zooming are applied to improve model generalization. A compact CNN architecture consisting of convolutional, pooling, flattening, and fully connected layers is designed to learn important features such as edges, shapes, symbols, and numbers from traffic-sign images. The model is trained using **backpropagation, gradient-based optimization, and cross-entropy loss**. Its performance is evaluated using **accuracy, precision, recall, F1-score, and a confusion matrix**. Sample predictions are also analyzed to identify classification errors, particularly among visually similar traffic-sign categories. The project demonstrates the practical application of CNNs in **computer vision and intelligent transportation systems**, with potential uses in driver-assistance systems, autonomous vehicles, road-safety monitoring, and smart transportation. It is aligned with **SDG 9 – Industry, Innovation and Infrastructure** and **SDG 11 – Sustainable Cities and Communities**.

Course Outcomes

- **CO1 – Learning Algorithms and Backpropagation:** Achieved through CNN training using backpropagation, gradient-based optimization, and cross-entropy loss.
- **CO2 – Designing Deep Architectures:** Achieved through the design and configuration of CNN layers for traffic-sign classification.
- **CO3 – Developing and Tuning CNNs for Vision Tasks:** Achieved through preprocessing, data augmentation, model training, parameter tuning, and evaluation using accuracy, precision, recall, F1-score, and confusion-matrix analysis

Guide Signature

Student Signature